

Intent-Driven Hierarchical DRL for Secrecy-Aware AoI–AoLI Optimization in RIS-Assisted HAP-IoT Communications

Muddassir Sadiq¹, Muhammad Sufyan Haider¹, Arooj Fatima¹, Muhammad Sohaib J. Solaija², *Member, IEEE*, Haejoon Jung³, *Senior Member, IEEE*, and Syed Ali Hassan⁴, *Senior Member, IEEE*

Abstract—The emergence of intent-based networking (IBN) has created new opportunities for Internet of Things (IoT) ecosystems to evolve from rigid, device-centric management toward artificial intelligence (AI)-native architectures capable of translating high-level intents into autonomous actions. In such systems, the dual requirements of information freshness and communication secrecy are critical, yet existing designs largely treat them in isolation. This article introduces an IBN-inspired hierarchical deep reinforcement learning (HDRL) framework for reconfigurable intelligent surface (RIS)-assisted millimeter-wave (mm-wave) IoT networks threatened by uncrewed aerial vehicle (UAV) eavesdroppers. The framework integrates a high-level RIS intent manager with a low-level power controller for transmit and jamming power adaptation, both trained via proximal policy optimization (PPO). A secrecy-gated transmission protocol further ensures that packets are withheld when confidentiality cannot be guaranteed. By embedding the tradeoff between age of information (AoI) and age of leaked information (AoLI) into the reward structure, the framework translates the high-level intent of *fresh yet secure* communication into context-aware, real-time network policies. Simulation results demonstrate significant reductions in AoI, improvements in AoLI, and enhanced secrecy efficiency under realistic fading and interference conditions. These findings underscore the potential of IBN-driven HDRL frameworks as foundational enablers for AI-powered, intent-

aware, and resilient IoT communication in beyond-5G and 6G networks.

Index Terms—Age of information (AoI), hierarchical deep reinforcement learning (HDRL), intent-based networking (IBN), Internet of Things (IoT), millimeter-wave (mm-wave), physical layer security (PLS), reconfigurable intelligent surface (RIS).

I. INTRODUCTION

THE rapid growth of Internet of Things (IoT) devices and mission-critical applications has shifted the paradigm of wireless communication networks toward low-latency, highly reliable, and context-aware architectures. Emerging services, such as smart grids, autonomous surveillance, industrial automation, and real-time healthcare monitoring, require timely and secure information delivery along with high data throughput. However, the growing heterogeneity and dynamic nature of these environments expose fundamental shortcomings in traditional device-centric and static architectures, necessitating more adaptive and autonomous designs. Therefore, intent-based networking (IBN) has gained prominence as it supports networks in translating high-level user intents, such as ensuring the freshness of sensor data while protecting against leakage, into dynamic, autonomous configurations, and policies.

Beyond conventional rule-based management, recent advances have extended IBN toward natural language-driven lifecycle management, where large language models (LLMs) decompose, translate, activate, and assure network intents, as demonstrated in real-world 5G deployments [1]. Similarly, distributed architectures such as 6G-INTENSE in [2] leverage IBN across all layers of 6G ecosystems, introducing abstractions like network–compute fabrics and pervasive artificial intelligence (AI) toolkits to manage multidomain heterogeneity. In next-generation IoT ecosystems, where dynamic adversaries, nonterrestrial links, and heterogeneous devices make static optimization impractical, such a design is especially appealing. Consequently, AI-native IBN frameworks, which combine programmable networking, big data, and machine learning to create truly self-optimizing systems, have become a key component of 6G and beyond IoT communications [3].

Despite these developments, maintaining information freshness and security in adversarial environments remains a critical

Received 15 September 2025; revised 27 November 2025; accepted 17 December 2025. Date of publication 29 December 2025; date of current version 27 April 2026. The work of Muhammad Sohaib J. Solaija and Syed Ali Hassan were supported in part by the North Atlantic Treaty Organization Science for Peace and Security (NATO-SPS) Program through the Project Shielding HAPS Against Intelligent Link-Level Attacks (SHAILLA) under Grant G8894. The work of Haejoon Jung was supported in part by MSIT, South Korea; in part by the National Research Foundation of Korea under Grant RS-2025-23323081; in part by the ITRC Support Programs under Grant IITP-2026-RS-2021-II212046; and in part by the Convergence Security Core Talent Training Business Support Program Supervised by IITP under Grant IITP-2023-RS-2023-00266615. (Corresponding authors: Haejoon Jung; Syed Ali Hassan.)

Muddassir Sadiq, Muhammad Sufyan Haider, and Arooj Fatima are with the School of Electrical Engineering and Computer Science (SEECs), National University of Sciences and Technology (NUST), Islamabad 44000, Pakistan (e-mail: msadiq.bee22seecs@seecs.edu.pk; mhaider.bee22seecs@seecs.edu.pk; arfatima.bee22seecs@seecs.edu.pk).

Muhammad Sohaib J. Solaija is with the Institute of Defence Technologies, Gebze Technical University, Gebze, 41400 Kocaeli, Türkiye (e-mail: solaija@ieeee.org).

Haejoon Jung is with the Department of Electronics and Information Convergence Engineering, Kyung Hee University, Yongin 17104, South Korea (e-mail: haejoonjung@khu.ac.kr).

Syed Ali Hassan is with SEECs, NUST, Islamabad 44000, Pakistan, and also with the Department of Electronic Engineering, Kyung Hee University, Yongin 17104, South Korea (e-mail: ali.hassan@seecs.edu.pk).

Digital Object Identifier 10.1109/IIOT.2025.3648959

2327-4662 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.

See <https://www.ieee.org/publications/rights/index.html> for more information.

challenge in IBN-enabled IoT. Classic performance metrics such as data throughput or packet error rate fall short in capturing the unique requirements of real-time IoT monitoring, where the age of information (AoI), the elapsed time since the most recently received update was generated, directly governs the quality of decision-making [4]. Unlike conventional cellular systems, where data rate and spectral efficiency dominate, IoT environments are energy-constrained, heterogeneous, and latency-sensitive, making AoI a more fundamental performance indicator [7]. In parallel, adversaries such as uncrewed aerial vehicle (UAV)-based eavesdroppers pose a threat of sensitive data leakage, making it essential to consider the age of leaked information (AoLI), a complementary metric that quantifies the staleness of information at the unintended receiver (UR) [5]. These insights position AoI–AoLI optimization as a cornerstone for intent-driven IoT systems, yet existing frameworks remain limited in their ability to balance timeliness and confidentiality under dynamic and adversarial conditions.

New physical-layer technologies such as reconfigurable intelligent surfaces (RISs) and millimeter-wave (mmWave) transmissions have attracted significant attention in order to achieve these goals in challenging wireless environments [6]. Through adaptive control of the phase shifts of passive reflecting elements, RIS offers a low-cost and energy-efficient way to reconfigure wireless propagation, making it especially suitable for low-power IoT devices. Beyond conventional beamforming gains, RIS-aided physical layer security (PLS) mechanisms can mitigate eavesdropping and jamming threats through resource allocation, cooperative communication, and artificial noise injection, thereby providing scalable protection for 6G-enabled IoT systems [6]. At the same time, mmWave bands offer large bandwidth for high-throughput IoT services but remain highly susceptible to blockage and severe path loss. Recent studies show that RIS-assisted mmWave architectures can counter these limitations by jointly optimizing transceiver beamforming, RIS reflection coefficients, and scheduling strategies, thereby improving system sum rate while satisfying stringent freshness constraints. Furthermore, when deployed between IoT devices and high-altitude platform (HAP) stations, RIS enables intent-driven beam steering that strengthens legitimate channels while weakening adversarial ones, while HAP-assisted networks extend coverage in rural and hard-to-reach regions and serve as elevated relays for synchronized jamming against aerial threats [8]. The promise of RIS-integrated nonterrestrial networks (NTNs) highlights the potential of these technologies to revolutionize beyond-5G and 6G connectivity [9]. However, their effectiveness in secrecy–freshness optimization depends on the coordinated design of multiple interdependent parameters, including sensor transmit power, HAP jamming power, and RIS phase configurations.

Conventional network optimization techniques such as convex programming and heuristic search have been widely used to address power control, beamforming, and secrecy in wireless networks [11]. However, these approaches generally rely on simplified channel models and static assumptions

that fail to capture the multiobjective and highly dynamic nature of nonterrestrial IoT systems. For instance, RIS-assisted mmWave communication frameworks have sought to improve timeliness by optimizing beamforming, RIS reflection coefficients, and scheduling strategies. Nevertheless, they remain constrained by heuristic decompositions and local searches that limit scalability. Similarly, alternating optimization techniques have been used to minimize AoI under aerial eavesdropping threats in UAV-assisted covert communication schemes. However, these solutions are still closely coupled to nonconvex formulations and depend on beamforming and carefully planned trajectories [10]. More system-level efforts, such as cooperative jamming-aided UAV-IoT networks, have attempted to jointly optimize secrecy outage probability (SOP) and AoI via Stackelberg game formulations, but still depend on strong channel state information (CSI) assumptions and often target single-objective tradeoffs [12]. Collectively, these works demonstrate that although tailored analytical techniques can yield encouraging improvements in timeliness or secrecy, they frequently come at the expense of generalizability and usefulness, highlighting the necessity for frameworks that can concurrently balance both objectives in the face of real-world uncertainties.

To overcome these challenges, the wireless community has increasingly turned to deep reinforcement learning (DRL), which enables agents to learn optimal transmission policies directly from environmental interaction rather than fixed analytical models. Recent studies demonstrate the promise of DRL in RIS-aided wireless systems, such as jointly optimizing beamforming and phase shifts under practical reflection amplitude models [13], and securing UAV–RIS networks against multiple eavesdroppers through policy-based DRL [14]. Beyond single-agent schemes, multiagent DRL has been proposed to design active and passive beam codebooks while drastically reducing training overhead [15]. Hierarchical DRL (HDRL) and advanced variants such as double deep Q networks (D3QNs) and soft-actor critic (SAC) have been leveraged to enhance secrecy and spectrum sharing in multi-IRS networks [16]. In parallel, DRL-driven approaches have been employed for AoI-aware scheduling, including RL-based peak AoI minimization in smart healthcare systems [17] and TD3-based UAV navigation for large-scale IoT data collection [18]. However, the majority of existing DRL-based works use a flat control architecture, in which a single agent controls every system parameter simultaneously. This results in poor scalability, slow convergence, and limited applicability to large RIS-assisted IoT networks.

Motivated by the multilayered decision processes characteristic of IBN, we propose an IBN-inspired HDRL framework [19] for RIS-assisted secure mmWave IoT systems. In the considered architecture, a high-powered IoT sensing device transmits updates toward a HAP over an RIS-assisted mmWave link, while a mobile UAV acts as an eavesdropper. To enhance secrecy, the HAP, operating in full-duplex mode, simultaneously generates jamming signals to disrupt the eavesdropper, and a secrecy-gated transmission protocol ensures that no data packets are transmitted whenever confidentiality cannot be guaranteed. Within this HDRL framework,

a high-level RIS manager configures long-term reflection policies, while a low-level power controller dynamically adjusts transmit and jamming powers on a per-slot basis. By embedding the dual objectives of minimizing AoI and maximizing AoLI into the reward structure, the network autonomously translates the high-level intent of “fresh and secure IoT communication” into real-time, context-aware actions, aligning with the vision of IBN-enabled AI-powered IoT networks.

While hierarchical DRL has been explored for RIS resource management in prior works such as [16] and [19], these studies primarily target metrics like secrecy rate, spectral efficiency, or generic throughput and do not model information freshness or leakage staleness as first-class design objectives. In contrast, our formulation embeds the *AoI–AoLI tradeoff directly into the IBN hierarchy*. Rather than simply adding AoI and AoLI as extra scalar terms in a flat DRL reward, we treat them as *high-level network intents* that shape the structure and coupling of the hierarchical controllers.

Concretely: 1) the high-level agent operates on aggregated AoI/AoLI statistics and long-term channel information to learn RIS reflection strategies that realize a desired freshness–secrecy balance; 2) the low-level agent observes instantaneous AoI/AoLI and CSI to translate this intent into fast transmit-jamming power adaptations; and 3) a secrecy-gated transmission policy ensures that AoI and AoLI influence not only the reward, but also whether updates are generated at all. This multitimescale, intent-driven coordination between RIS configuration and power control is absent in existing HDRL-based RIS frameworks [16], [19], where hierarchical structure is used mainly to decompose rate or secrecy optimization.

To the best of our knowledge, this is the first work to employ AoI and AoLI as semantic drivers of hierarchical control in an IBN-inspired RIS-assisted HAP–UAV system, enabling a unified and scalable approach to timeliness and confidentiality in adversarial IoT networks. Furthermore, the proposed intent-driven HDRL design is directly applicable to practical 6G IoT scenarios such as disaster-response monitoring via HAP–IoT links, privacy-preserving smart surveillance, and industrial sensing deployments where timely yet confidential status updates are critical. Based on this design, our main contributions are summarized below.

- 1) We propose an IBN-inspired HDRL framework that integrates the AoI–AoLI tradeoff as a high-level network intent, jointly driving RIS configuration and power control for secure IoT communication, which is not captured by existing HDRL-based RIS resource management schemes [16], [19].
- 2) We develop an RIS-assisted HAP uplink model over mmWave channels with Nakagami fading and UAV eavesdroppers, incorporating realistic pathloss and reflection effects.
- 3) We formulate a secrecy-aware AoI–AoLI optimization problem considering sensor transmit power, HAP jamming with residual self-interference (RSI), and discrete RIS phase shifts.
- 4) We introduce a multitimescale hierarchical control design, where the HLC updates RIS configurations at

coarse intervals while the LLC performs fine-grained slot-wise power adaptation, enabling scalable and efficient HDRL operation.

- 5) We validate the proposed framework through extensive simulations, demonstrating reduced AoI, increased AoLI, and improved secrecy efficiency compared to baseline DRL and non-DRL schemes.
- 6) In response to scalability and deployment concerns, we provide a detailed computational complexity analysis for all DRL and hierarchical DRL algorithms considered, including proximal policy optimization (PPO), SAC, TD3, H-PPO, and H-SAC. The analysis highlights the sublinear growth enabled by RIS phase grouping and the reduced per-episode cost achieved through hierarchical decomposition.

The remainder of this article is organized as follows. Section II presents the system model. Section III formulates the secrecy-aware AoI–AoLI optimization problem and introduces the proposed HDRL framework. Section IV describes the simulation environment and experimental setup. Section V presents and discusses the system performance and simulation results. Finally, Section VI concludes the article and outlines directions for future research.

II. SYSTEM MODEL

We consider an uplink IoT system consisting of a ground-based source node S , a RIS with N reflecting elements, a HAP represented by H serving as the legitimate receiver, and a mobile UAV U acting as an eavesdropper (see Fig. 1). The system is slotted, with each time slot of duration T_s during which the channels are assumed to remain constant. The RIS provides programmable reflections to enhance the desired link and suppress the eavesdropper, while the HAP operates in full-duplex mode, simultaneously receiving information and emitting artificial noise to confuse the UAV.

A. Channel and Path Loss Model

The large-scale path loss between any two nodes i and j , separated by a distance d_{ij} (in km), is modeled as follows:

$$L_{ij} [\text{dB}] = 92.45 + 20 \log_{10}(f_{\text{GHz}}) + 20 \log_{10}(d_{ij}) + \alpha d_{ij} + \zeta_{ij} \quad (1)$$

where f_{GHz} is the carrier frequency in GHz, α denotes the atmospheric attenuation per kilometer, and ζ_{ij} represents an additional shadowing loss term (notably higher for UAV-related links). The linear-scale gain is $L_{ij}^{-1} = 10^{-L_{ij}/10}$.

The small-scale fading on each link is modeled as Nakagami- m , yielding the equivalent baseband channel

$$h_{ij} = \sqrt{L_{ij}^{-1}} \tilde{h}_{ij}, \quad \tilde{h}_{ij} \sim \text{Nakagami}(m_{ij}) \quad (2)$$

where m_{ij} represents the fading severity parameter corresponding to the channel connecting nodes i and j .

The thermal noise power at each receiver over bandwidth B is given by the following equation:

$$\sigma^2 = N_0 B \cdot 10^{\text{NF}/10} \quad (3)$$

where N_0 is the one-sided noise power spectral density and NF is the receiver noise figure in dB.

B. RIS-Assisted Channels

The RIS introduces a controllable phase shift to each impinging signal. Let g_n denote the channel from the source S to the n th RIS element, and $u_{k,n}$ the channel from the n th RIS element to node $k \in \{H, U\}$. The reflection coefficient of the n th element is

$$\phi_n = \eta_{\text{RIS}} e^{j\theta_n} \quad (4)$$

where $\eta_{\text{RIS}} \in (0, 1]$ is the amplitude efficiency and θ_n is phase shift chosen from a finite set.

The composite reflected channel from the RIS to node k is expressed as follows:

$$h_{Rk} = \sum_{n=1}^N g_n \phi_n u_{k,n} \quad (5)$$

and the effective end-to-end channels for HAP and UAV are, respectively, given by the following equations:

$$h_{\text{eq},H} = h_{\text{SH}} + h_{\text{RH}} \quad (6)$$

$$h_{\text{eq},U} = h_{\text{SU}} + h_{\text{RU}} \quad (7)$$

where h_{SH} and h_{SU} denote the direct channels from the source to the HAP and UAV, respectively, while h_{RH} and h_{RU} represent the RIS-assisted components. Thus, $h_{\text{eq},H}$ corresponds to the overall equivalent channel gain between the source and the HAP, and $h_{\text{eq},U}$ corresponds to the overall equivalent channel gain between the source and the UAV.

Including the antenna gains and hardware losses at the transmitter and receivers, the effective power gains toward HAP and UAV are modeled as follows:

$$G_H = |h_{\text{eq},H}|^2 G_t G_H^{(\text{rx})} \quad (8)$$

$$G_U = |h_{\text{eq},U}|^2 G_t G_U^{(\text{rx})} \quad (9)$$

where G_t denotes the transmit antenna gain, and $G_H^{(\text{rx})}$ and $G_U^{(\text{rx})}$ denote the receive gains (including hardware losses) at the HAP and UAV, respectively.

C. Artificial Jamming Links

The HAP, which is in full duplex mode, emits artificial noise with power P_j to confuse the eavesdropper. Let h_{JH} and h_{JU} denote the effective self-interference and jamming channels toward the HAP and UAV, respectively. The associated power gains are

$$G_{\text{JH}} = |h_{\text{JH}}|^2, \quad G_{\text{JU}} = |h_{\text{JU}}|^2 \quad (10)$$

where G_{JH} incorporates additional attenuation and a RSI factor at the HAP, and G_{JU} incorporates a possible jamming gain boost toward the UAV.

D. Signal Model, SINR, and Secrecy Rate

At each time slot, the source allocates its total transmit power P_{tot} between the information-bearing signal power P_t and the cooperative jamming power P_j , subject to

$$\begin{aligned} P_{\text{tot}} &= P_t + P_j \\ \text{s.t. } P_{\text{min}} &\leq P_{\text{tot}} \leq P_{\text{max}}, \quad P_t \geq 0, \quad P_j \geq 0. \end{aligned} \quad (11)$$

The received baseband signals at the HAP and UAV can be written as follows:

$$y_H = \sqrt{P_t} H x_s + \sqrt{\kappa_H P_j} h_{\text{JH}} x_j + n_H \quad (12)$$

$$y_U = \sqrt{P_t} U x_s + \sqrt{P_j} h_{\text{JU}} x_j + n_U \quad (13)$$

where x_s and x_j denote the information and jamming symbols with unit power, $\kappa_H \in (0, 1]$ is the RSI factor at the HAP, and n_H, n_U are AWGN terms with variance σ^2 .

Therefore, the received signal-to-interference-plus-noise ratio (SINR) at the HAP is given by the following equation:

$$\gamma_H = \frac{P_t |H|^2}{\sigma^2 + \kappa_H P_j G_{\text{JH}}} \quad (14)$$

and the SINR at the UAV is expressed as follows:

$$\gamma_U = \frac{P_t |U|^2}{\sigma^2 + P_j G_{\text{JU}}}. \quad (15)$$

Using (14) and (15), the achievable transmission rates for HAP and UAV are given by the following equations:

$$R_H = B \log_2(1 + \gamma_H) \quad (16)$$

$$R_U = B \log_2(1 + \gamma_U) \quad (17)$$

where B represents the system bandwidth, and the instantaneous secrecy capacity is defined as follows:

$$C_s(t) = [R_H(t) - R_U(t)]^+. \quad (18)$$

E. Age-of-Information and Age-of-Leaked-Information

The information freshness at the HAP is quantified using AoI. Let $\Delta_H(t)$ denote the AoI at time slot t . Its evolution is expressed as follows:

$$\Delta_H(t+1) = \begin{cases} t - g_H(t), & \mu_H(t) = 1 \\ \Delta_H(t) + 1, & \text{otherwise} \end{cases} \quad (19)$$

where $g_H(t)$ denotes the generation time of the packet delivered at the HAP in slot t .

Similarly, the AoLI at the UAV, denoted by $\Delta_U(t)$, evolves as follows:

$$\Delta_U(t+1) = \begin{cases} t - g_U(t), & \mu_U(t) = 1 \\ \Delta_U(t) + 1, & \text{otherwise} \end{cases} \quad (20)$$

where $g_U(t)$ represents the generation time of the packet intercepted at the UAV. Thus, $\Delta_H(t)$ captures the freshness of legitimate updates at the HAP, while $\Delta_U(t)$ reflects the staleness of intercepted information at the eavesdropper.

III. SECURITY-AWARE AOI AND AOI OPTIMIZATION

A. Optimization Problem Formulation

The overarching design goal is to guarantee secure communication in the considered IoT system, where the legitimate HAP receives timely updates while the eavesdropping UAV is only exposed to stale information. To achieve this, we jointly optimize three tightly coupled aspects: 1) adaptively allocating source transmit and jamming power to prevent information leakage; 2) minimizing the AoI at the HAP for data freshness;

and 3) maximizing the AoLI at the UAV to enforce secrecy through staleness.

The long-term average AoI $\bar{\Delta}_H$ at the HAP over a horizon of T slots is expressed as follows:

$$\bar{\Delta}_H = \frac{1}{T} \sum_{t=1}^T \Delta_H(t) \quad (21)$$

where T denotes the total number of time slots in the observation window. On the other hand, the long-term average AoLI $\bar{\Delta}_U$ at the UAV is defined as follows:

$$\bar{\Delta}_U = \frac{1}{T} \sum_{t=1}^T \Delta_U(t). \quad (22)$$

Here, reducing $\bar{\Delta}_H$ ensures the timely delivery of legitimate updates, whereas increasing $\bar{\Delta}_U$ guarantees that any information leaked to the UAV is outdated and hence less useful. The secrecy requirement is enforced through the instantaneous secrecy capacity constraint, which is expressed as follows:

$$C_s(t) \geq R_s \quad \forall t. \quad (23)$$

If this condition is not satisfied, the source remains idle, thereby preventing potential leakage.

The joint optimization problem is thus formulated as follows:

$$\min_{\{P_t(t), P_j(t), \phi(t)\}} \bar{\Delta}_H - \omega \bar{\Delta}_U \quad (24a)$$

$$\text{s.t.} \quad P_{\min} \leq P_t(t) + P_j(t) \leq P_{\max} \quad (24b)$$

$$P_t(t), P_j(t) \geq 0 \quad \forall t \quad (24c)$$

$$C_s(t) \geq R_s \quad \forall t \quad (24d)$$

$$\theta_n \in \mathcal{F} \quad \forall n \quad \forall t \quad (24e)$$

where $\omega > 0$ is a tunable parameter balancing timeliness against secrecy staleness, $\phi(t)$ denotes the RIS phase shift configuration, and $\mathcal{F}$ represents the discrete RIS phase shift set. Despite the clarity of the above formulation, directly solving the joint optimization problem is highly intractable. The coupling between source transmit power, jamming power, and discrete RIS phase shifts makes the problem nonconvex, while the stochastic fading, UAV mobility, and secrecy constraints render conventional optimization ineffective. Moreover, the conflicting objectives of minimizing $\bar{\Delta}_H$ and maximizing $\bar{\Delta}_U$ introduce a multiobjective tradeoff that cannot be handled by static or model-based schemes.

To overcome these challenges, we resort to a learning-based solution. While DRL has shown promise for adaptive IoT communications, flat control architectures suffer from poor scalability and slow convergence. Inspired by the hierarchical decision structure of IBN, we propose a HDRL framework that separates RIS phase adaptation by a high-level agent and dynamic power allocation by a low-level agent. This design enables efficient, scalable, and real-time translation of the high-level intent of fresh yet secure IoT communication into practical transmission strategies.

TABLE I
SIMULATION PARAMETERS

Parameter	Value
Carrier frequency f	28 GHz
Bandwidth B	20 MHz
Slot duration T_s	1 s
S $\rightarrow$ RIS distance	10 m
S $\rightarrow$ UAV distance	5 km
S $\rightarrow$ HAP distance	20 km
RIS $\rightarrow$ HAP distance	20 km
RIS $\rightarrow$ UAV distance	5 km
RIS elements N	64
RIS efficiency η	0.9
Transmit and Jamming power range	0–30 dBm
Residual SI factor κ	0.35
Atmospheric loss	0.06 dB/km
UAV shadowing margin	6 dB
Antenna gains/losses	Source: 8/1 dB; HAP: 20/1 dB; UAV: 1/1.5 dB
Nakagami- m params	$m_{SR}=2, m_{RH}=3, m_{SH}=3, m_{RU}=1.5, m_{SU}=2$
Noise figure	5 dB
Frame size	2 Mbits (1.5M payload + 0.5M redundancy)
DRL discount γ	0.99
GAE λ	0.95
HLC rollout length	128 steps
LLC rollout length	2048 steps
Episodes $\times$ steps	100 episodes $\times$ 10,000 slots

B. Proposed HDRL Framework

To implement our HDRL framework, each base DRL algorithm $\mathcal{A} \in \{\text{PPO}, \text{SAC}, \text{TD3}\}$, we construct a hierarchical agent denoted by H- $\mathcal{A}$, consisting of a HLC for RIS phase adaptation and a LLC for power allocation. Both controllers interact with the *same* environment and share a common scalar reward signal r_t , but they maintain *separate* function approximators and experience buffers. Hence, there is no direct gradient sharing between the HLC and LLC; instead, their interaction is mediated through the environmental dynamics and the state features that each level observes. Fig. 2 shows the proposed framework.

At the low level, the LLC acts at every slot t , producing a continuous action

$$a_t^L = (P_t, P_j)$$

from the state s_t^L (which contains slot-level CSI, AoI/AoLI, and power-related features). This yields transitions

$$(s_t^L, a_t^L, r_t, s_{t+1}^L)$$

which are stored in a low-level experience buffer. For the on-policy case (H-PPO), these transitions form rollouts of length L_{LL} that are used once to update the policy/value networks. For the off-policy cases (H-SAC and H-TD3), the transitions are accumulated in a replay buffer $\mathcal{B}_L$ from which mini-batches are sampled to perform critic and actor updates according to the underlying algorithm $\mathcal{A}$.

The HLC operates on a slower timescale with period K slots (see Table I). At high-level decision index k , it observes a slow-timescale state s_k^H (containing averaged AoI/AoLI statistics and large-scale channel summaries) and outputs an action a_k^H corresponding to a RIS phase configuration. This configuration is then held fixed while the LLC executes for the next K slots. The high-level controller receives an aggregated return

$$\tilde{r}_k = \sum_{t=kK}^{(k+1)K-1} \gamma^{t-kK} r_t$$

and we store high-level transitions as follows:

$$(s_k^H, a_k^H, \tilde{r}_k, s_{k+1}^H).$$

As in the low level, H-PPO uses these trajectories in an on-policy manner with generalized advantage estimation (GAE), while H-SAC and H-TD3 store them in a separate replay buffer $\mathcal{B}_H$ and apply their respective off-policy updates to the HLC networks.

Thus, the coupling between HLC and LLC is realized in two complementary ways: 1) the LLC's slot-wise actions determine the immediate secrecy performance and AoI/AoLI evolution, which in turn shape the aggregated reward $\tilde{r}_k$ used to train the HLC; and 2) the HLC's RIS configuration modifies the effective channel gains, which appear in the LLC observation vector (e.g., via the normalized RIS index and channel gains) and thereby influence the low-level policy updates.

To mitigate nonstationarity arising from this multitimescale structure, we employ a warm-start schedule. During the first N_{warmup} episodes (parameter `warmup_episodes` in the simulator), only the LLC is updated while the HLC policy is kept fixed. After this warm-up, both levels are trained jointly, with the LLC updated more frequently (every L_{LL} slots) and the HLC updated every K slot. This schedule stabilizes learning and provides clearer gradient signals for the high-level agent across all hierarchical variants (H-PPO, H-SAC, and H-TD3).

1) *Reward Function*: A unified reward signal captures the tradeoff between secrecy, timeliness, and power efficiency

$$r(t) = -\Xi\Delta_H(t) + \beta\Delta_U(t) - \lambda_t P_t(t) - \lambda_j P_j(t). \quad (25)$$

where $\Delta_H(t)$ and $\Delta_U(t)$ denote the AoI and AoLI increments, respectively, and $\Xi, \beta > 0$ balance their relative importance. Power penalties λ_t, λ_j regulate energy consumption. To enforce confidentiality, a large negative reward is imposed when the instantaneous secrecy capacity $C_s(t)$ falls below the target secrecy rate R_s .

2) *Key Advantages*: Unlike monolithic DRL, the proposed HDRL exploits system hierarchy to

- 1) Exploit the distinct timescales of RIS configuration and power control.
- 2) Reduce training sample complexity.
- 3) Accelerate convergence in high-dimensional environments.
- 4) Ensure timely and secure communication under adversarial jamming.

IV. SIMULATION ENVIRONMENT

A. System Topology and Channel Settings

The IoT source is located on the ground, the RIS panel is deployed in the vicinity of the source, the UAV eavesdropper hovers 5 km from the source, and the HAP receiver operates at an altitude of 20 km. The RIS is equipped with 64 passive reflecting elements with reflection efficiency $\eta = 0.9$. The transmission frequency is set to 28 GHz with 20 MHz bandwidth, and the slot duration is fixed at $T_s = 1$ s. Source transmit power and HAP jamming power are varied in the range from 0 to 30 dBm.

Algorithm: H-PPO for RIS Phase and Power Control

- 1: **Initialize**: Environment $\mathcal{E}$, HLC policy π_H , LLC policy π_L , value networks V_H, V_L
- 2: **Input**: Horizon T , update interval K , learning rates (η_H, η_L)
- 3: **for** each episode **do**
- 4: Reset $\mathcal{E}$; obtain initial states (s_H, s_L)
- 5: **for** slot $t = 1, \dots, T$ **do**
- 6: **if** $t \bmod K = 1$ **then**
- 7: HLC selects RIS phase $a_H \sim \pi_H(s_H)$
- 8: Apply a_H to configure RIS
- 9: **end if**
- 10: LLC selects power allocation $a_L = (P_t, P_j) \sim \pi_L(s_L)$
- 11: Execute (a_H, a_L) in $\mathcal{E}$; observe reward

$$r_t = -\Xi\Delta_H(t) + \beta\Delta_U(t) - \lambda_t P_t - \lambda_j P_j$$

and next states (s'_H, s'_L)

- 12: Store $(s_H, a_H, s_L, a_L, r_t)$ in buffers $\mathcal{B}_H, \mathcal{B}_L$
- 13: Update $(s_H, s_L) \leftarrow (s'_H, s'_L)$
- 14: **end for**
- 15: Estimate advantages with GAE:

$$\hat{A}_t = \sum_{l=0}^{\infty} (\gamma\lambda)^l (r_{t+l} + \gamma V(s_{t+l+1}) - V(s_{t+l}))$$

- 16: Compute policy ratio for PPO:

$$r_t(\theta) = \frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$$

- 17: Optimize clipped surrogate loss:

$$L^{\text{PPO}}(\theta) = \mathbb{E}_t[\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t)]$$

- 18: Add entropy bonus $\mathcal{H}[\pi_{\theta}]$ and value loss:

$$L(\theta) = L^{\text{PPO}}(\theta) + c_v(R_t - V(s_t))^2 - c_e \mathcal{H}[\pi_{\theta}]$$

- 19: Update $\theta_H \leftarrow \theta_H + \eta_H \nabla_{\theta} L(\theta_H)$
 - 20: Update $\theta_L \leftarrow \theta_L + \eta_L \nabla_{\theta} L(\theta_L)$
 - 21: **end for**
 - 22: **Output**: Trained HLC (RIS phase policy) and LLC (power allocation policy)
-

Propagation losses include free-space path loss (FSPL) with atmospheric absorption of 0.06 dB/km. UAV links incur an additional 6 dB shadowing margin. Small-scale fading follows a Nakagami- m distribution with m parameters set to $(m_{\text{SR}}, m_{\text{RH}}, m_{\text{SH}}, m_{\text{RU}}, m_{\text{SU}}) = (2.0, 3.0, 3.0, 1.5, 2.0)$. Thermal noise is calculated as $N_0 = -174 + 10 \log_{10}(B) + \text{NF}$ dBm with noise figure $\text{NF} = 5$ dB. Antenna configurations are: source $G_t = 8$ dBi/ $L_t = 1$ dB, HAP $G_{\text{hap}} = 20$ dBi/ $L_{\text{hap}} = 1$ dB, and UAV $G_{\text{uav}} = 1$ dBi/ $L_{\text{uav}} = 1.5$ dB.

B. Traffic and Transmission Protocol

Each time slot, the source generates one 2 Mbit update frame, divided into 1.5 Mbit payload and 0.5 Mbit redundancy header. A secrecy-gated transmission protocol is implemented:

if the instantaneous secrecy capacity is below the target payload rate, a dummy packet is transmitted to prevent leakage. The HAP also emits jamming signals toward the UAV, with RSI modeled by $\kappa = 0.35$.

C. Baseline Algorithms

To provide a comprehensive and fair evaluation of the proposed HDRL framework, we compare its performance against a diverse set of baselines, including conventional DRL agents, their hierarchical extensions, and an analytical optimization method. The baselines are summarized as follows.

- 1) *Random Policy*: RIS phases and power levels (P_t, P_j) are sampled uniformly at random in every slot.
- 2) *PPO*: A single-agent PPO jointly optimizes RIS phases and (P_t, P_j) within a monolithic action space.
- 3) *SAC*: A single-agent SAC is employed to handle the continuous action space associated with joint RIS and power control.
- 4) *TD3*: A single-agent TD3 performs deterministic continuous control over transmit power, jamming power, and RIS phases using twin-critic stabilization.
- 5) *H-PPO*: A hierarchical PPO design where the high-level controller adjusts RIS configurations over coarse time intervals, and the low-level controller performs slot-wise power adaptation.
- 6) *H-SAC*: A hierarchical SAC structure that separates long-term RIS control from fast-timescale power decisions while leveraging entropy-regularized exploration.
- 7) *H-TD3*: A hierarchical TD3 variant that divides RIS phase selection and joint power allocation into two decision layers to improve scalability.
- 8) *Analytical AO Baseline*: To complement the learning-based baselines and address the need for an analytical benchmark, we also consider a static alternating-optimization (AO) design under simplified, tractable assumptions. Specifically, we assume: 1) perfect instantaneous CSI for all links (S-HAP, S-UAV, S-RIS, RIS-HAP, RIS-UAV); 2) deterministic large-scale channel gains without small-scale fading randomness; 3) a single common RIS phase shift ϕ applied to all reflecting elements; and 4) a fixed total power budget P_{tot} split as $P_t = sP_{\text{tot}}$ and $P_j = (1-s)P_{\text{tot}}$ with a scalar power-split factor $s \in [0, 1]$.

Using the SINR and rate expressions R_H and R_U already defined in the system model, we introduce the following scalar utility to capture the tradeoff between improving the legitimate link and degrading the eavesdropper

$$U(s, \phi) = \alpha R_H(s, \phi) - \Lambda R_U(s, \phi) \quad (26)$$

where $\alpha > 0$ and $\Lambda > 0$ are design weights that reflect the relative importance of AoI reduction (via higher R_H) and AoLI enhancement (via lower R_U).

Since $U(s, \phi)$ is not jointly convex in (s, ϕ) , we adopt a standard AO procedure widely used in RIS-based secrecy optimization. In each AO iteration, we first fix s and optimize the RIS phase shift by the following equation:

$$\phi^*(s) = \arg \max_{\phi \in [0, 2\pi)} U(s, \phi)$$

which can be implemented via a 1-D search over the feasible phase grid and then fix $\phi = \phi^*(s)$ and optimize the scalar power split by the following equation:

$$s^*(\phi) = \arg \max_{0 \leq s \leq 1} U(s, \phi)$$

using a simple line search over s . These two steps are alternated until convergence or a maximum number of iterations is reached. The resulting pair (s^*, ϕ^*) yields a closed-form-free but fully analytical RIS-power configuration, which we report as the *AO baseline* alongside the hierarchical DRL schemes H-PPO, H-SAC, and H-TD3. These baselines span naive random control, standard DRL agents, hierarchical multitimescale architectures, and an analytical optimizer, providing a rigorous comparison against the proposed HDRL framework. Comparative performance trends of all baselines are discussed in Section V.

D. HDRL Training Hyperparameters

The proposed HDRL framework integrates an HLC for RIS configuration and an LLC for transmit/jamming power control. Both controllers adopt PPO, implemented with multilayer perceptrons (MLPs) consisting of two hidden layers (128 and 256 units) with ReLU activations. The following hyperparameters govern stability and performance during training.

- 1) *Discount Factor*: $\gamma = 0.99$ ensures that long-term rewards, such as sustained freshness (AoI) and secrecy (AoLI), are considered while still emphasizing immediate slot-level gains.
- 2) *GAE*: parameter $\lambda = 0.95$ balances bias and variance in policy gradient estimation, allowing the agent to stabilize learning under highly variable fading channels.
- 3) *Entropy Coefficient*: 10^{-3} promotes policy exploration by penalizing premature convergence to suboptimal deterministic strategies, which is crucial for discovering robust RIS and power control policies in dynamic UAV mobility scenarios.
- 4) *Gradient Clipping*: maximum norm of 0.5 prevents unstable weight updates that could otherwise occur due to the large variance of secrecy-driven rewards.

Temporal granularity is managed through hierarchical decision intervals. The HLC updates RIS reflection phases every K slots (rollout length = 128), reflecting the slower adaptation timescale of hardware reconfiguration. Conversely, the LLC acts at every slot (rollout length = 2048), dynamically adjusting transmit and jamming powers in response to instantaneous channel variations.

Training is conducted for 100 episodes, each episode containing up to 10000 time slots, enabling convergence under diverse channel realizations and UAV trajectories. Throughout training, the environment logs per-episode averages of AoI, AoLI, transmit power, and jamming power to a CSV file, providing reproducible metrics for comparative evaluation. Table I summarizes these simulation parameters.

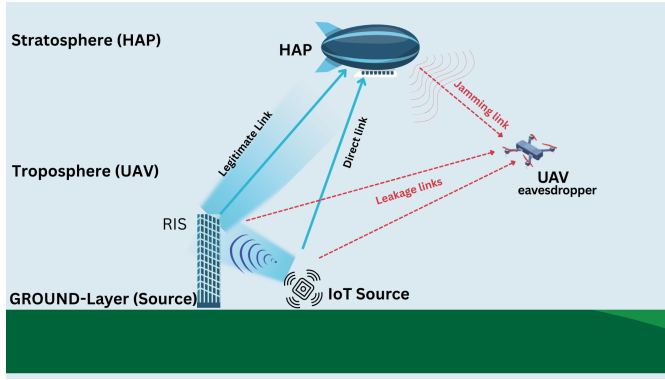


Fig. 1. RIS-assisted mmWave IoT uplink with HAP and UAV eavesdropper.

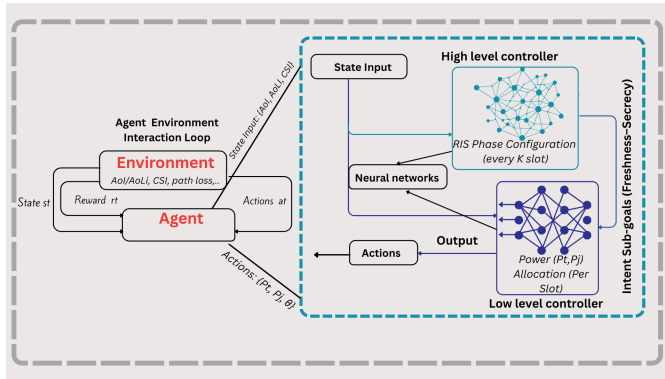


Fig. 2. Illustration of the hierarchical policy structure and agent-environment interaction loop.

E. Sensitivity Analysis

To quantify the robustness of the framework, we varied key system parameters, including the number of RIS elements, UAV distance, and jamming power. Results indicate that AoI improves monotonically with larger RIS arrays, while AoLI benefits more strongly from increased jamming power. However, excessive jamming beyond 30 dBm reduces AoI gains due to RSI. UAV mobility analysis revealed a performance threshold: secrecy deteriorates sharply when UAV distance drops below 3 km, underscoring the importance of adaptive HDRL-based policies.

V. RESULTS AND DISCUSSION

We will analyze the convergence behavior of AoI and AoLI for both simple and hierarchical models. The goal is to evaluate the timeliness of updates and the staleness of intercepted information under different policy optimization strategies, and to highlight how the hierarchical architecture improves convergence efficiency. The convergence behaviors of simple architecture for AoI and AoLI metrics are shown in Fig. 3 and for hierarchical architecture are shown in Fig. 4 for AoI, and in Figs. 5 and 6 for AoLI.

A. Convergence of AoI for Simple and Hierarchical Models

In the case of simple models, such as SAC, PPO, TD3, random, and AO, AoI fluctuates significantly over time and

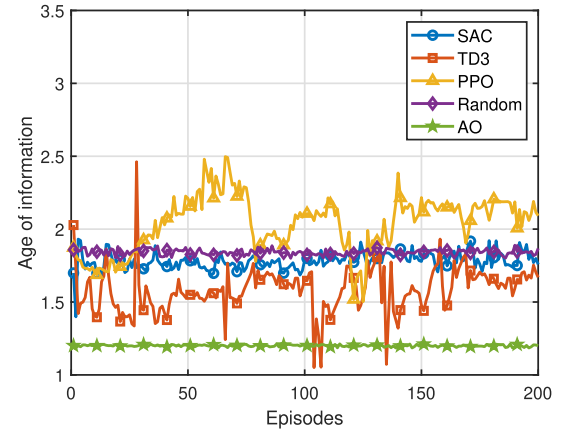


Fig. 3. AoI convergence for flat architecture.

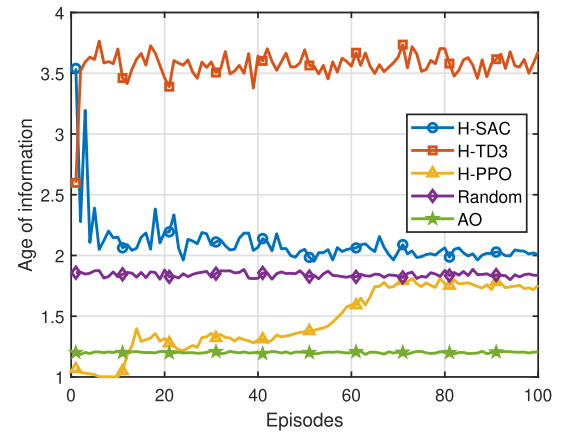


Fig. 4. AoI convergence for hierarchical architecture.

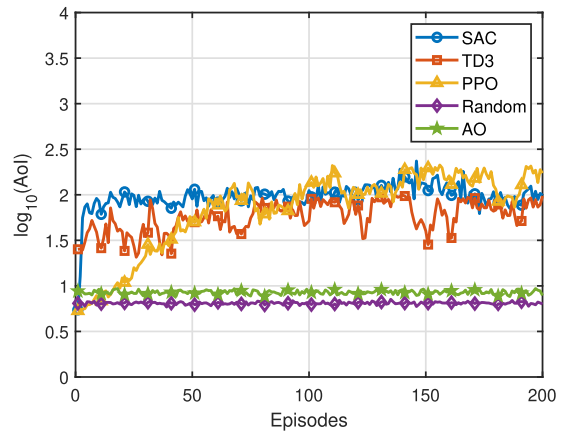


Fig. 5. AoLI convergence for flat architecture.

does not properly converge to a stable value. As shown in Fig. 3, AoI exhibits high variability, indicating that these models struggle to find an optimal balance between information freshness and efficiency.

In contrast, for the hierarchical models H-SAC, H-PPO, and H-TD3, the convergence of AoI is much smoother. As seen in Fig. 4, all hierarchical models converge to a stable AoI value within just 20 episodes. This fast convergence demonstrates that the hierarchical structure, by dividing decision-making into high-level and low-level controllers,

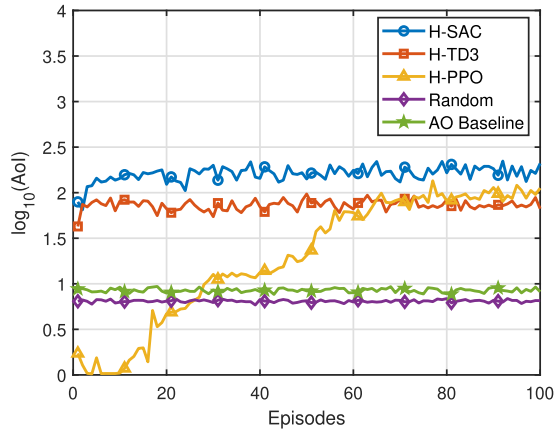


Fig. 6. AoLI convergence for hierarchical architecture.

significantly improves the model's ability to optimize AoI efficiently.

B. Convergence of AoLI for Flat and Hierarchical Models

For AoLI, both the flat and hierarchical models show similar convergence behaviors. As seen in Figs. 5 and 6, the fluctuations in AoLI are comparable between the two model types. This indicates that while the general trends in AoLI are similar for both simple and hierarchical models, the difference lies in the computational efficiency. The hierarchical models are computationally more efficient, as discussed in Section V-C.

C. Computational Complexity and Inference Time

After analyzing the convergence behavior of AoI and AoLI for both simple and hierarchical models, we now turn our attention to the computational complexity of the algorithms. In this section, we evaluate the per-episode computational complexity of the models and discuss the efficiency of the hierarchical structure in reducing training costs and inference times.

1) *Computational Complexity of Hierarchical Models:* The hierarchical models (H-PPO, H-SAC, and H-TD3) show significant improvements in computational efficiency compared to their flat counterparts (PPO, SAC, and TD3). The primary advantage of the hierarchical structure lies in the separation of decision-making into high-level and low-level controllers, which allows for sublinear growth in complexity as the number of RIS elements increases.

a) *Computational complexity of H-PPO:* The per-episode computational complexity of the proposed H-PPO scales linearly with the number of low-level steps L and is dominated by the forward and backward passes of the actor-critic networks at both hierarchical levels. The complexity is given by $\mathcal{O}(4L(h_{LL}^2 + (h_{HL}^2/M)))$, where h_{LL} and h_{HL} represent the hidden layer sizes of the low-level and high-level networks, respectively, and M is the interval between successive high-level decisions. The term h_{LL}^2 captures the cost of the LLC optimizing (P_t, P_j) at each step, while h_{HL}^2/M reflects the cost of updating the RIS configuration by the HLC.

The factor of 4 accounts for both the forward and backward computations of the actor and critic networks. Thanks to this hierarchical structure, the complexity grows sublinearly with the number of RIS elements N .

b) *Computational complexity of H-SAC:* The per-episode computational complexity of the proposed H-SAC scales linearly with the number of low-level steps L and is dominated by the forward and backward passes of the actor and twin-critic networks at both hierarchical levels. The complexity for H-SAC is given by $\mathcal{O}(6L(h_{LL}^2 + (h_{HL}^2/M)))$. The factor 6 accounts for the computations of the actor, the two critics, and the temperature parameter update in SAC. The term h_{LL}^2 captures the per-step cost of the LLC that optimizes (P_t, P_j) , while h_{HL}^2/M corresponds to the amortized cost of HLC that updates the RIS phase configuration every M steps. The complexity grows sublinearly with the number of RIS elements N , as h_{HL} scales with the number of RIS phase groups $G \ll N$, ensuring that H-SAC is computationally more efficient than flat SAC.

c) *Computational complexity of H-TD3:* For the H-TD3, the per-episode computational complexity scales linearly with the number of low-level steps L and is dominated by the forward and backward passes of the actor and twin Q -networks at both hierarchical levels. The complexity is expressed as $\mathcal{O}((4 + (2/D))L(h_{LL}^2 + (h_{HL}^2/M)))$, where D is the delay between actor updates and M is the interval between successive high-level decisions. The factor $(4 + (2/D))$ accounts for the forward and backward computations of the critics and the amortized actor update. Similar to H-PPO and H-SAC, the complexity grows sublinearly with the number of RIS elements N by grouping the RIS phases into smaller groups, thus reducing the training cost significantly.

2) *Computational Complexity of Flat Models:* For comparison, we also discuss the computational complexity of the flat models, namely PPO, SAC, and TD3. These models do not benefit from hierarchical decomposition, leading to a direct increase in complexity as the number of RIS elements increases.

a) *Computational complexity of PPO:* The per-episode computational complexity of the PPO agent scales linearly with the episode length W and the number of gradient updates T , and is dominated by the forward and backward passes of the actor and critic networks. The complexity for PPO is expressed as $\mathcal{O}(W(1 + T)(h_{\pi}^2 + h_V^2 + d_a))$, where h_{π} and h_V denote the hidden layer widths of the actor and critic networks, respectively, and d_a represents the action dimension, which includes the RIS phase configuration and power control variables (P_t, P_j) . Unlike hierarchical models, PPO's complexity scales directly with the number of RIS elements N , making it less efficient for large-scale deployments.

b) *Computational complexity of SAC:* Similar to PPO, the per-episode computational complexity of SAC scales linearly with the episode length W and the number of gradient updates per step T , and is dominated by the forward and backward passes of the actor, two Q -networks, and the value network. The overall complexity is expressed as $\mathcal{O}(W(1 + T)(h_{\pi}^2 + 2h_Q^2 + h_V^2 + d_a))$. Here, h_{π} , h_Q , and h_V represent the hidden-layer widths of the actor, critic, and value networks,

TABLE II

PER-EPISODE COMPUTATIONAL COMPLEXITY AND INFERENCE TIME OF DRL AND HIERARCHICAL DRL ALGORITHMS

Algorithm	Per-Episode Complexity	Inference Time (ms)
H-PPO	$\mathcal{O}(4L(h_{LL}^2 + \frac{h_{HL}^2}{M}))$	0.022536476
H-TD3	$\mathcal{O}(4L(h_{LL}^2 + \frac{h_{HL}^2}{M}))$	0.144205561
H-SAC	$\mathcal{O}(6L(h_{LL}^2 + \frac{h_{HL}^2}{M}))$	0.150215268
PPO	$\mathcal{O}(W(1+T)(h_{\pi}^2 + h_V^2 + d_a))$	0.048018098
SAC	$\mathcal{O}(W(1+T)(h_{\pi}^2 + 2h_Q^2 + h_V^2 + d_a))$	0.168781757
TD3	$\mathcal{O}(W(1+T)(h_{\pi}^2 + 2h_Q^2 + d_a))$	0.160237303

respectively, and d_a is the action dimension. SAC's complexity also scales directly with the total number of RIS elements N , leading to increased computational cost as the RIS size grows.

c) *Computational complexity of TD3*:: The per-episode computational complexity of TD3 grows linearly with the episode length W and the number of gradient updates per step T , and is primarily dominated by the forward and backward passes through the policy (actor) and twin Q-networks (critics). The complexity is given by $\mathcal{O}(W(1+T)(h_{\pi}^2 + 2h_Q^2 + d_a))$, where h_{π} and h_Q denote the hidden-layer widths of the actor and critic networks, respectively. Similar to PPO and SAC, the action dimension d_a grows with the number of RIS elements N , making TD3 less computationally efficient for large-scale systems compared to hierarchical models.

3) *Inference Time Analysis*: In addition to the training complexity, inference time is another important factor to consider, especially in real-time applications. Inference time is measured in milliseconds (ms), representing the time taken for a single model evaluation during deployment. From Table II, we observe that hierarchical models (H-PPO, H-TD3, and H-SAC) generally have lower inference times compared to the flat models (PPO, SAC, and TD3). This further demonstrates the advantage of the hierarchical architecture, which leads to faster decision-making during deployment. The hierarchical models consistently outperform their flat counterparts in terms of both training complexity and inference time. By leveraging a two-level structure, where the high-level controller focuses on RIS configuration and the low-level controller handles transmit and jamming power optimization, the hierarchical models ensure that the growth in computational complexity remains sublinear with respect to the number of RIS elements. This makes hierarchical models not only faster in training but also more efficient in real-time applications, making them ideal for large-scale IoT deployments.

Among all considered methods, H-PPO achieves the lowest per-step inference time while maintaining the smallest per-episode complexity. Therefore, unless otherwise stated, the subsequent performance results for the hierarchical architecture are reported using H-PPO as the representative method.

D. Impact of RIS Configuration

Fig. 7 illustrates the impact of increasing the number of RIS elements on the achieved AoI at the HAP and AoLI at

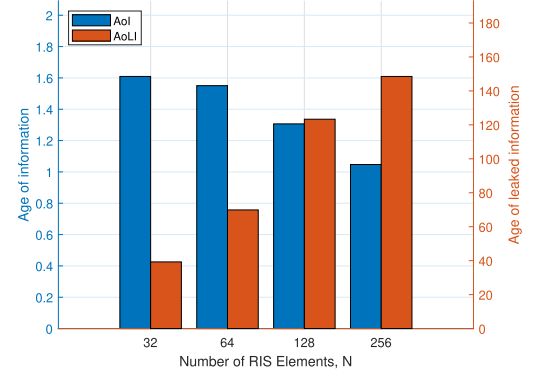


Fig. 7. AoI and AoLI versus number of RIS elements N .

the UAV. As the number of meta elements of RIS grows from $N = 32$ to $N = 256$, the AoI consistently decreases, indicating that larger RIS apertures provide stronger beamforming gain and more reliable uplink delivery, thereby improving the timeliness of legitimate updates. In contrast, the AoLI increases sharply with N , reflecting that the UAV receives progressively more outdated information. This behavior results from the enhanced RIS-assisted signal focusing toward the HAP, coupled with the reduced effective channel strength at the UAV. For $N = 256$, the AoLI becomes almost an order of magnitude larger than in the $N = 32$ case, while the AoI simultaneously attains its lowest value. These results highlight an important observation that enlarging the RIS primarily strengthens secrecy by degrading the eavesdropper's ability to obtain fresh updates, while improving freshness at the legitimate receiver. This demonstrates the effectiveness of RIS scaling in simultaneously enhancing timeliness and confidentiality within the proposed HDRL framework. While larger RIS arrays clearly enhance freshness and secrecy, this gain comes at the cost of increased hardware complexity and longer DRL training time due to the higher dimensional control space. Fig. 8 characterizes the influence of the RIS array size and the sensor-RIS distance d_{SR} on the secrecy capacity and the eavesdropper rate. We see a clear trend as the number of RIS elements increases C_s grows monotonically for all values of d_{SR} . This improvement stems from the enhanced passive beamforming gain provided by larger RIS surfaces, which strengthens the effective composite channel toward the HAP and improves its received SINR. In contrast, the R_u exhibits an overall decreasing trend with N , particularly for smaller d_{SR} , indicating that RIS scaling not only boosts the legitimate link but also reduces the UAV's ability to decode the uplink signal.

The impact of d_{SR} is equally significant. When the sensor is closer to the RIS ($d_{SR} = 4$ m), the reflected link is strong, resulting in the highest secrecy rate and the fastest degradation of R_u . As d_{SR} increases to 6 and 9 m, both the secrecy performance and the eavesdropper suppression weaken due to the increased path loss before reflection, reducing the overall beamforming gain at the HAP and causing the UAV's received signal quality to deteriorate less aggressively. These results collectively demonstrate that secrecy performance depends jointly on the available RIS aperture and its

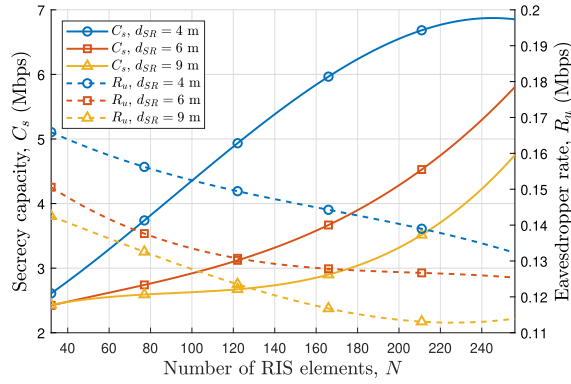


Fig. 8. Secrecy capacity C_s and eavesdropper rate R_u versus the number of RIS elements N for different sensor–RIS distances d_{SR} .

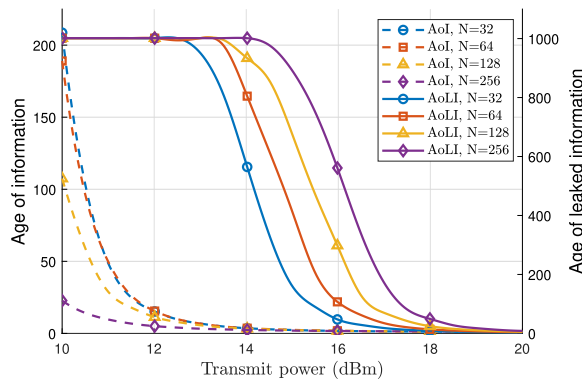


Fig. 9. AoI and AoLI versus transmit power for different numbers of RIS elements N .

geometric placement: larger RIS arrays and shorter sensor–RIS distances produce the strongest constructive combining at the HAP while most effectively diminishing the eavesdropper’s decoding capability.

E. Impact of Transmit Power on Freshness–Secrecy Tradeoff

Fig. 9 illustrates the evolution of AoI and AoLI with respect to the P_t for different numbers of meta-elements of RIS. As P_t increases, the legitimate link strengthens, and the AoI at the HAP rapidly decreases once the received SINR crosses the decoding threshold. However, further increasing P_t yields diminishing returns because the AoI eventually reaches its minimum achievable value. A similar pattern is observed for AoLI, although larger RIS arrays delay the eavesdropper’s ability to decode, as seen earlier in Fig. 7, a sufficiently high P_t inevitably boosts the UAV’s SINR as well, allowing it to receive fresher updates and thereby reducing AoLI. This creates an inherent tradeoff that increasing P_t improves AoI but simultaneously erodes secrecy by lowering AoLI. Therefore, optimal operation requires jointly tuning P_t and P_j so that the HAP maintains a strong link for timely updates while the UAV remains jammed into a high-AoLI regime. For the static UAV case examined in Fig. 9, the optimal operating range is approximately $P_t = 14$ – 16 dBm, where the AoI has already reached its minimum value while the AoLI remains high before its sharp decline. RIS size influences the shape of

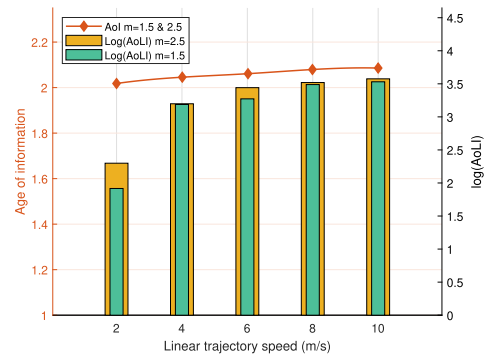


Fig. 10. AoI and AoLI versus linear trajectory with varying m .

these curves as shown in Fig. 7, but the dominant factor is the balance between transmit power and jamming power needed to manage the AoI–AoLI tradeoff.

F. Robustness of the HDRL Policy Under Stochastic and Dynamic UAV Mobility

To assess the robustness of the proposed HDRL framework under dynamic and stochastic UAV mobility, we evaluate performance under two representative motion patterns. The first is a linear random-walk trajectory, and the second is a circular trajectory with varying angular speeds. These mobility patterns introduce distinct forms of spatial variation in the UAV–HAP and UAV–RIS channels and allow us to test how well the trained HDRL policy generalizes to previously unseen movement dynamics.

1) *Linear Random-Walk Mobility*: In the first scenario, the UAV moves with a constant linear velocity within each time slot while updating its direction stochastically after every episode. The direction is selected from the set {North, South, East, West} according to a discrete uniform distribution, where each direction is chosen with probability 0.25. This induces random fluctuations in both the reflected and direct eavesdropping channels. To further evaluate robustness to fading variations, we analyze two Nakagami- m conditions, $m \in \{1.5, 2.5\}$, applied to the RIS–UAV and sensor–UAV links. As shown in Fig. 10, the AoI remains nearly constant across speeds and fading conditions, increasing only modestly as the UAV moves faster and channel coherence time decreases. In contrast, AoLI exhibits stronger dependence on fading severity: for $m = 2.5$, the UAV benefits from a more reliable channel and achieves lower AoLI, while for $m = 1.5$, the severe fading combined with high mobility significantly degrades the eavesdropper’s ability to decode, leading to consistently higher AoLI. These results confirm that the HDRL policy maintains low AoI while preserving high AoLI under stochastic linear motion.

2) *Circular Trajectory Mobility*: In the second scenario, the UAV follows a circular trajectory with speed varying from 1 to 9 m/s. This motion creates periodic channel fluctuations as the UAV periodically approaches and moves away from the dominant scattering path. Fig. 11 illustrates that AoI again increases only mildly with speed for $m = 2.5$, indicating that the HDRL agent adapts effectively to the cyclic variations in

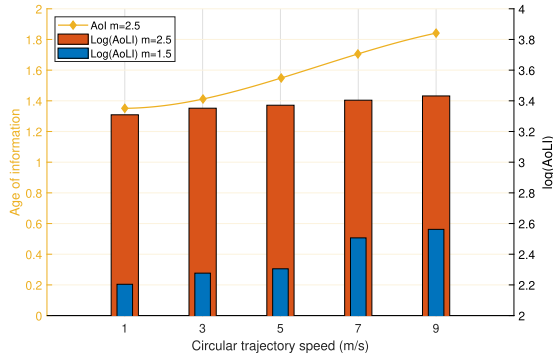


Fig. 11. AoI and AoLI versus circular trajectory with varying m .

channel quality. More prominently, AoLI remains significantly higher for $m = 1.5$ across all speeds due to the combined effect of severe fading and reduced interception opportunity as the UAV rapidly sweeps through its trajectory. For $m = 2.5$, AoLI decreases slightly at lower speeds but increases again as velocity grows, reflecting the reduced time spent near strong eavesdropping positions.

Overall, the combined results from the linear and circular mobility models demonstrate that the proposed HDRL framework is highly robust to stochastic, dynamic, and geometry-dependent UAV motion. The learned policy generalizes beyond the static training environment, consistently achieving low AoI and high AoLI even when the UAV exhibits random or continuous high-speed movement, thereby addressing reviewer concerns regarding cross-environment robustness, fading variability, and realistic UAV mobility behavior. While the proposed HDRL framework improves secrecy and freshness, its scalability is affected by increased computational overhead when the RIS size and state space grow. Larger network models require more memory and training time, which makes them challenging to deploy in real time on resource-constrained IoT devices. Future work can focus on lightweight HDRL solutions, such as phase-grouping, model compression, and edge-assisted learning, to reduce complexity and enable practical large-scale implementation.

VI. CONCLUSION

This article proposed an HDRL framework for intent-driven AoI–AoLI optimization in RIS-assisted HAP IoT communications under UAV eavesdropping. Experimental results demonstrated that the framework significantly improves information freshness, secrecy efficiency, and resilience compared to baseline methods. However, the computational complexity of hierarchical training grows with the number of RIS elements and environmental states, which may challenge real-time scalability in large-scale IoT deployments. Future research should focus on developing lightweight HDRL schemes, extending the framework to multi-RIS and intelligent UAV scenarios, and validating performance on hardware testbeds for practical nonterrestrial IoT networks.

REFERENCES

- [1] A. Mekrache, A. Ksentini, and C. Verikoukis, "Intent-based management of next-generation networks: An LLM-centric approach," *IEEE Netw.*, vol. 38, no. 5, pp. 29–36, Sep. 2024.
- [2] A. Boutouchent et al., "6G-INTENSE: Intent-driven native artificial intelligence architecture supporting network-compute abstraction and sensing at the deep edge," *IEEE Veh. Technol. Mag.*, vol. 20, no. 1, pp. 44–54, Mar. 2025.
- [3] J. Yang, X. Tang, Y. Zhou, L. Liu, and W. Jiangzhou, "6G native intelligence network architecture enabled by intent abstraction and knowledge," *J. Commun.*, vol. 44, no. 2, pp. 1–14, 2023.
- [4] İ. Kahraman, A. Köse, M. Koca, and E. Anarim, "Age of information in Internet of Things: A survey," *IEEE Internet Things J.*, vol. 11, no. 6, pp. 9896–9914, Mar. 15, 2024.
- [5] L. Zheng, J. Ren, Y. Liu, and Q. Chen, "Analysis and optimization of age of information and age of leaked information for IoT networks," *IEEE Wireless Commun. Lett.*, vol. 13, no. 12, pp. 3583–3587, Dec. 2024.
- [6] W. Khalid, M. A. U. Rehman, T. Van Chien, Z. Kaleem, H. Lee, and H. Yu, "Reconfigurable intelligent surface for physical layer security in 6G-IoT: Designs, issues, and advances," *IEEE Internet Things J.*, vol. 11, no. 2, pp. 3599–3613, Jan. 2024.
- [7] M. Sufyan Haider, M. Sadiq, A. Fatima, N. Mazhar, H. Jung, and S. Ali Hassan, "DRL-based AoI-driven scheduling for EH-enabled CR-NOMA IoT networks with queue constraints," *IEEE Wireless Commun. Lett.*, vol. 15, pp. 470–474, 2026.
- [8] W. U. Khan, A. Mahmood, C. K. Sheemar, E. Lagunas, S. Chatzinotas, and B. Ottersten, "Reconfigurable intelligent surfaces for 6G non-terrestrial networks: Assisting connectivity from the sky," *IEEE Internet Things Mag.*, vol. 7, no. 1, pp. 34–39, Jan. 2024.
- [9] A. Kumar et al., "Reconfigurable intelligent surface (RIS)-assisted non-terrestrial network (NTN)-based 6G communications: A contemporary survey," *Sensors*, vol. 24, no. 21, p. 6958, Oct. 2024.
- [10] S. Salar Hosseini, P. Azmi, and A. Nazari, "Age of information minimization in UAV-assisted covert communication: Trajectory and beamforming design," 2024, *arXiv:2411.12312*.
- [11] W. Wu and B. Wang, "Robust secrecy beamforming for wireless information and power transfer in multiuser MISO communication system," *EURASIP J. Wireless Commun. Netw.*, vol. 2015, no. 1, Dec. 2015, Art. no. 161.
- [12] Y. Ma, K. Liu, Y. Liu, and L. Zhu, "Timeliness and secrecy-aware uplink data aggregation for large-scale UAV-IoT networks," *IEEE Internet Things J.*, vol. 11, no. 10, pp. 17341–17356, May 2024.
- [13] B. Saglam, D. Gurgunoglu, and S. S. Kozat, "Deep reinforcement learning based joint downlink beamforming and RIS configuration in RIS-aided MU-MISO systems under hardware impairments and imperfect CSI," in *Proc. IEEE Int. Conf. Commun. Workshops (ICC Workshops)*, Rome, Italy, May 2023, pp. 66–72.
- [14] A. Iqbal, A. Al-Habashna, G. Wainer, F. Bouali, G. Boudreau, and K. Wali, "Deep reinforcement learning-based resource allocation for secure RIS-aided UAV communication," in *Proc. IEEE 98th Veh. Technol. Conf. (VTC-Fall)*, Oct. 2023, pp. 1–6.
- [15] A. Abdallah, A. Celik, M. M. Mansour, and A. M. Eltawil, "Multi-agent deep reinforcement learning for beam codebook design in RIS-aided systems," *IEEE Trans. Wireless Commun.*, vol. 23, no. 7, pp. 7983–7999, Jul. 2024.
- [16] L. Wang, W. Wu, F. Zhou, Q. Wu, O. A. Dobre, and T. Q. S. Quek, "Hybrid hierarchical DRL enabled resource allocation for secure transmission in multi-RIS-assisted sensing-enhanced spectrum sharing networks," *IEEE Trans. Wireless Commun.*, vol. 23, no. 6, pp. 6330–6346, Jun. 2024.
- [17] B. Wu, Z. Cai, W. Wu, and X. Yin, "AoI-aware resource management for smart health via deep reinforcement learning," *IEEE Access*, vol. 11, pp. 81180–81195, 2023.
- [18] J. Zhang, K. Kang, M. Yang, H. Zhu, and H. Qian, "AoI-minimization in UAV-assisted IoT network with massive devices," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Austin, TX, USA, Apr. 2022, pp. 1290–1295.
- [19] M. Umer, M. A. Mohsin, A. A. Nasir, H. Abou-Zeid, and S. A. Hassan, "Intelligent spectrum sharing in integrated TN-NTNs: A hierarchical deep reinforcement learning approach," *IEEE Wireless Commun.*, vol. 32, no. 3, pp. 64–71, Jun. 2025.